

World 8 - Z0-A Final Architecture Baseline

Provider-independent persistent runtime with independent observation and proposal-only development

Status: FINAL DESIGN BASELINE / NOT PRODUCTION

Frozen design date: 2026-08-24

Scope: Normative Z0-A architecture. Implementation and evidence promotion remain separately gated.

Central rule	Cognition and interfaces are replaceable; accountable identity, canonical state, authority, objectives, and accepted history are not.
Fixed topology	Five planes: Canonical Spine, Operational, Observation, Development/Mason, Evidence/Governance.
Mason boundary	Proposal only. No direct change to Objective, Hard Constraints, Authority, Identity, Canonical Spine, or Promotion.
Claim boundary	This document freezes the Z0-A design. It does not claim production readiness, autonomous evolution, or full autonomy.

1. Final architectural ruling

World 8 Z0-A is a persistent, auditable runtime whose source of truth remains outside replaceable model providers, sessions, user interfaces, telemetry systems, and development agents. Every accepted change must move from a known canonical state to a new canonical state through explicit authority, commit-time CAS, intent-bound idempotency, append-oriented history, and claim-specific evidence.

2. The five fixed planes

Plane	Owns	Must not become
Canonical Spine	World/Entity identity, event history, head/version, checkpoints, authorizations, canonical revisions	A model/UI/telemetry-controlled store
Operational	Entity/Role/RoleBinding, Task, ingress, Brain Gateway, context, outbox/effects/connectors	A provider-owned identity domain
Observation	Raw telemetry, normalization, aggregation, SLO evaluation, gap signals	Canonical truth or a Mason-controlled detector
Development / Mason	Candidate Phenotype Revision proposals	Self-authorizing/self-promoting evolution
Evidence / Governance	Proof Registry, evaluator receipts, approvals, policy, promotion gates, audit	A mechanism that inherits evidence between unrelated claims

3. Entity, Role, and RoleBinding

Entity and Role are distinct. **company-001** is an operational Entity/Society. **secretary-role** is a responsibility slot activated through a governed **role_binding**. Provider/model/session identifiers are provenance, not canonical object identity.

4. Versioned Objective Contract and Phenotype Revision

The Objective Contract is a canonical, versioned contract containing objective, success metrics/SLOs, hard constraints, authority ceiling, risk class, observation contract, allowed adaptation surface, and promotion policy. A Phenotype Revision is a versioned candidate execution change referencing its parent revision and objective version, with explicit migration, rollback, metrics, risk, and evidence.

Objective Contract -> constrains Observation and allowed adaptation
Phenotype Revision -> proposes a reversible runtime change
Constitutional change -> separate governance path for Objective/Hard Constraints/Authority/Identity

5. Observation precedes Mason

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raw telemetry -> normalization -> aggregation/windowing -> SLO/invariant evaluation ->
gap_signal -> Evidence Registry -> Mason may consume
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Mason may not select the evidence set, redefine the SLO, or manufacture a gap. Gap detection may be deterministic or may use an independently versioned classifier, but that detector remains outside Mason and must be independently testable.

6. Mason is proposal-only

- Mason references a valid gap and current Objective Contract version.
- It proposes a minimal reversible change with tests and rollback.
- It cannot modify Objective, Hard Constraints, Authority, Identity, or Canonical Spine.
- It cannot evaluate or promote its own candidate.
- It cannot directly write canonical state.

7. Independent evaluator identity

Database constraints prove allowed database capabilities, not the real identity/version of the evaluator that produced a result. Each Evaluation Receipt therefore binds evaluator principal, role/credential reference, evaluator build digest, candidate revision, objective version, test-suite/environment digest, result metrics, evidence references, and attestation/signature where available. Mason, evaluator, sequencer, effect executor, and promotion authority use separate capabilities and credentials.

8. Canonical Spine commit contract

- **Commit-time CAS:** expected head is checked within the transaction that advances the head.
- **Intent-bound idempotency:** identical retry deduplicates; key reuse with changed semantics is a collision error.
- **Atomic append/head/outbox:** event append, head advance, permitted projections, and planned effects are transactionally coupled.
- **Transactional outbox + receipts:** effects are obligations; attempts and provider-side receipts settle them. Only scoped effectively-once semantics are targeted.
- **Sequencer lease + fencing:** stale/displaced sequencers cannot commit.
- **DB-enforced append-only history:** runtime roles cannot UPDATE/DELETE/TRUNCATE committed events.
- **Verified replay:** chain, hashes, versions, duplicate IDs, and final head are checked before materialization.

- **Clean-host restore:** reconstruction must work from durable artifacts and accept one fresh governed post-restore write under a fresh lease/fencing token.

9. Provider-independent continuity and scoped context

A provider switch must preserve the same World/Entity/Society/Role/Task IDs and canonical lineage. Context loading is scoped K0-K4: World kernel, Society/Entity context, Role contract, current Task, and minimum provenance-bearing retrieved evidence.

10. Full runtime and development path

External input

- > Operational ingress/auth/route
- > scoped context + replaceable cognition
- > proposal/result
- > authority/policy evaluation
- > Canonical commit
- > task/state projection + optional outbox
- > effect attempt -> receipt
- > Observation telemetry
- > SLO evaluation -> gap_signal
- > Mason candidate revision
- > independent evaluation
- > Governance promotion decision
- > canonical revision activation

11. Evidence and governance discipline

Every claim carries its own predicate, falsifier, required evidence level, status, and evidence references. OPEN means required evidence does not exist. Neighboring PASS results do not upgrade an unrelated claim.

Level	Meaning
E0	Defined / untested
E1	Static contract/schema/privilege evidence
E2	Local/disposable executable evidence
E3	Persistent/integration evidence
E4	Production-like restore/security/load/chaos/independent credentials
E5	Longitudinal evidence under predefined SLO/failure envelope

12. Separation of duties

Capability	Allowed	Forbidden
Promotion Authority / Human Root	Approve/revoke gated promotion	Rewrite history or replace security controls
Sequencer	Commit with valid lease/fencing/CAS	Self-approve
Observer	Measure and emit attributed signals	Canonical mutation or objective definition
Mason	Propose candidate revision	Self-evaluate or promote
Evaluator	Judge candidate and emit receipt	Propose the same candidate or promote it
Effect Executor	Settle authorized obligations	Create its own authorization

13. Promotion and rollback

1. Observation records a valid gap.
2. Mason or a human proposer creates a candidate revision.
3. Independent shadow/replay/test evaluation runs.
4. An attributed Evaluation Receipt is committed.
5. Governance verifies hard constraints and authority are unchanged.
6. Promotion Authority decides.
7. Revision activation is committed with CAS against the current canonical revision/head.
8. Regression triggers a new rollback event to a known-good revision.

14. Explicit non-claims

- No claim of consciousness, biological life, AGI, or legal personhood.
- Mason is not proven autonomous evolution in Z0-A.
- Hash chains are tamper-evident, not privileged-storage security.
- Database constraints alone do not prove evaluator identity.
- Human approval alone is not a sufficient security boundary.
- No universal exactly-once external-effects claim.
- Production readiness remains gated on credential separation, persistent-target restore, security/load/chaos testing, and SLOs.

15. Z0-A exit gates

Gate	PASS criterion
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Plane isolation	Five planes enforced in APIs/schema/DB roles with bypass tests.
Entity/Role split	Portable RoleBinding; provider metadata not required for canonical identity.
Objective/Revision versioning	Immutable/versioned contracts with CAS activation.
Observation independence	raw->aggregate->SLO->gap outside Mason with provenance.
Mason confinement	Negative tests prove it cannot change hard boundaries or promote.
Evaluator identity	Receipts bind principal/build/input/output and credentials are separated.
Spine atomicity	CAS/idempotency/append/head/outbox/fencing adversarial gates PASS.
Recovery	Clean-host restore plus fresh governed post-restore write PASS.
Effects	Authorization->outbox->attempt->receipt and duplicate/ambiguity tests PASS.

16. Lineage from World 8 v0.1.1

World 8 v0.1/v0.1.1 remains important operational evidence for world-001, company-001, RoleBindings, Task Bus, and persistent canonical history. Z0-A closes the next architectural layer by fixing the five-plane model, separating Observation from Mason, versioning Objective/Phenotype Revision, and making evaluator identity evidence explicit. Prior evidence does not automatically PASS these new Z0-A gates.